CO2-AT3- Debugging and Optimization

Name: Ajay Kumar Reddy

Reg.no :192425274

Course: CSA0712 - COMPUTER NETWORKS

Faculty: Dr. S. SALINI

1. The Half-Duplex Bottleneck

- **Scenario: A corporate user on PC_A reports slow network speeds and random connection drops during large file transfers to a local server. Both devices are connected to a 2960 switch (Switch_A).**
- **Packet Tracer Check: Open the CLI on Switch_A and issue show interfaces FastEthernet 0/1. You notice a rapidly increasing count of late collisions and runts, alongside a Half-duplex status. The PC is hardcoded to Full-duplex.**
- **Debugging Question: Identify the root cause of the frame corruptions. State the exact command used to discover this issue on the switch interface, and explain how to resolve it inside Cisco Packet Tracer.**

2. The Native VLAN Mismatch

- **Scenario: You connect Switch_1 and Switch_2 via their G0/1 ports. Host devices on VLAN 10 attached to Switch_1 cannot ping host devices on VLAN 10 attached to Switch_2.**
- **Packet Tracer Check: Looking at the CLI console logs, periodic Cisco Discovery Protocol (CDP) warning alerts flood the screen.**
- **Debugging Question: Interpret the console message %CDP-4-NATIVE_VLAN_MISMATCH. Which command will show you the operational native VLAN trunking status of G0/1 on both switches? Fix the issue so that VLAN 10 traffic successfully passes across the trunk line.**

4. The Blocked Port Mystery

- **Scenario: You build a redundant topology using three switches connected in a triangle to prevent network failure. Host PC_1 pings Host PC_2, but you notice the initial ICMP packet drops, and the link light on one of the switch ports stays amber.**
- **Packet Tracer Check: Switch to Simulation Mode. Filter traffic by STP and ICMP. Watch the Spanning Tree Protocol Bridge Protocol Data Units (BPDUs) cycle.**
- **Debugging Question: Use the command show spanning-tree to determine why the amber port is in a BLK (Blocking) state. Is this a network error or intended behavior? Explain your answer using the concept of layer 2 loops.**

1. The Half-Duplex Bottleneck

Root Cause

The problem is caused by a **duplex mismatch**.

- **PC_A** is configured for **Full-Duplex**.
- **Switch_A (FastEthernet0/1)** is operating in **Half-Duplex**.
- When one device uses Full-Duplex and the other uses Half-Duplex, the half-duplex side detects **late collisions**, **runts**, and frame corruption, resulting in slow network performance and intermittent connection drops.

Command Used to Discover the Issue

On **Switch_A**, enter:

Switch_A# show interfaces FastEthernet0/1

Typical output:

FastEthernet0/1 is up, line protocol is up

Hardware is Fast Ethernet

MTU 1500 bytes

Full/Half Duplex: Half-duplex

Late collisions: 45

Runts: 28

The important indicators are:

- **Half-duplex** status
- Increasing **Late collisions**
- Increasing **Runts**

These confirm a duplex mismatch.

Resolution in Cisco Packet Tracer

Method 1 (Recommended)

Enable **Auto Negotiation** on both devices.

On the PC:

PC → Config → FastEthernet

Duplex → Auto

On the switch:

Switch_A(config)# interface FastEthernet0/1

Switch_A(config-if)# duplex auto

Switch_A(config-if)# speed auto

Method 2

Configure **both devices** for **Full-Duplex**.

Switch_A(config)# interface FastEthernet0/1

Switch_A(config-if)# duplex full

Switch_A(config-if)# speed 100

Also configure the PC network adapter to:

Speed : 100 Mbps

Duplex : Full

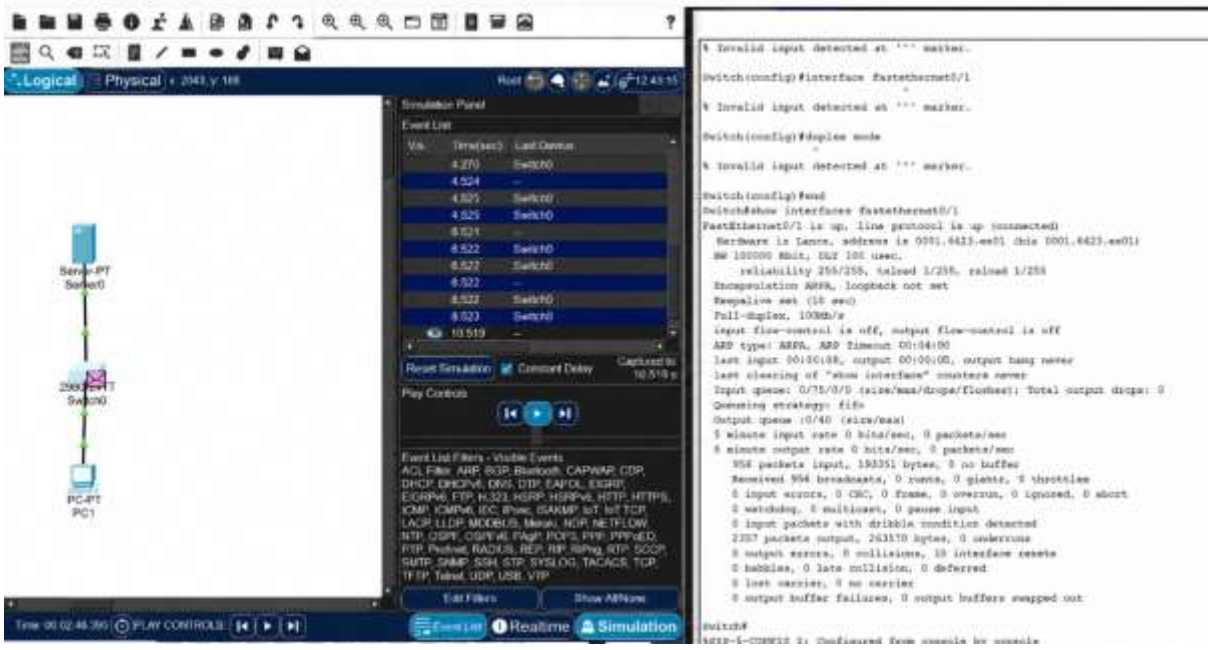
Verification

Run:

Switch_A# show interfaces FastEthernet0/1

Expected result:

- Full-duplex
- No increase in late collisions
- No increase in runts
- Stable and faster communication



2. The Native VLAN Mismatch

Root Cause

The error message

%CDP-4-NATIVE_VLAN_MISMATCH

indicates that the **native VLAN configured on one end of the trunk link is different from the native VLAN configured on the other end**.

Example:

- Switch_1 G0/1 → Native VLAN **10**
- Switch_2 G0/1 → Native VLAN **1**

Because the native VLANs do not match, untagged traffic is placed into different VLANs, preventing VLAN 10 hosts from communicating correctly.

Command to Check the Native VLAN

On both switches, run:

Switch# show interfaces trunk

Typical output:

Port Mode Encapsulation Status Native vlan

Gi0/1 on 802.1q trunking 10

Compare the **Native vlan** value on both switches.

You can also verify the interface configuration using:

Switch# show running-config interface GigabitEthernet0/1

Resolution in Cisco Packet Tracer

Configure the same native VLAN on both trunk ports.

Example (using VLAN 10):

Switch_1

Switch_1(config)# interface GigabitEthernet0/1

Switch_1(config-if)# switchport mode trunk

Switch_1(config-if)# switchport trunk native vlan 10

Switch_2

Switch_2(config)# interface GigabitEthernet0/1

Switch_2(config-if)# switchport mode trunk

Switch_2(config-if)# switchport trunk native vlan 10

If VLAN 10 does not exist, create it first:

Switch(config)# vlan 10

Switch(config-vlan)# name SALES

Verification

Run:

Switch# show interfaces trunk

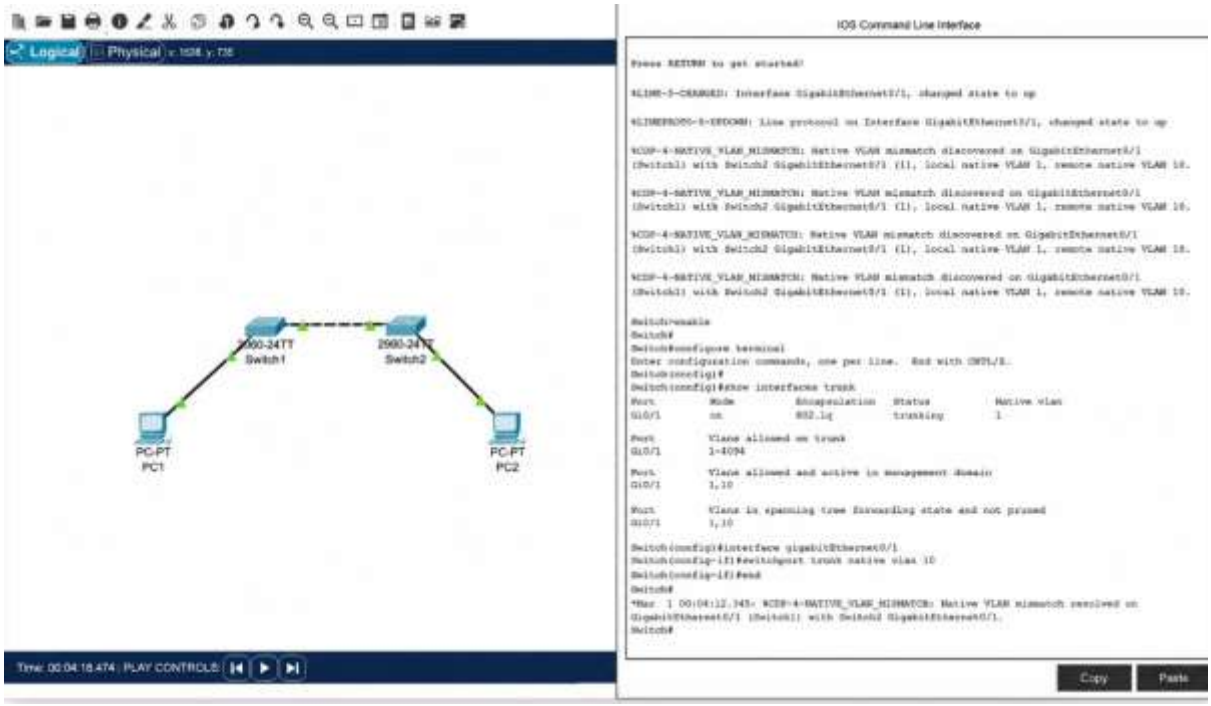
Expected output:

Port Mode Status Native vlan

Gi0/1 on trunking 10

Then test connectivity:

PC> ping <IP address of the host in VLAN 10 on Switch_2>



4. The Blocked Port Mystery

Root Cause

The network consists of **three switches connected in a triangle topology**, creating multiple physical paths between the switches. These redundant links provide backup connectivity but also introduce the possibility of a **Layer 2 loop**. To prevent this condition, **Spanning Tree Protocol (STP)** automatically analyzes the network topology and selects a single active path for data transmission. One of the redundant ports is placed into the **Blocking (BLK)** state, which is indicated by an **amber link light** in Cisco Packet Tracer. This behavior is expected and is part of the normal STP operation.

Command Used to Verify the STP Status

On the switch CLI, execute the following command:

Switch# show spanning-tree

Example output:

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769

Address 000A.F3B1.123

Bridge ID Priority 32769

Address 000D.BD45.5678

Interface Role Sts Cost

Gi0/1 Root FWD 4

Gi0/2 Desg FWD 4

Gi0/3 Altn BLK 4

The output displays the STP roles and states assigned to each interface. The **Root Port** and **Designated Port** remain in the **Forwarding (FWD)** state, while the **Alternate Port** is placed in the **Blocking (BLK)** state because it represents the redundant path.

Analysis

The show spanning-tree command confirms that the switch has successfully completed the STP election process. The network has selected a **Root Bridge**, and every switch has calculated the shortest path to reach the Root Bridge. Since multiple paths are available, STP blocks one redundant port to eliminate switching loops. As a result, data traffic is forwarded only through the active forwarding ports while the blocked port remains available as a backup link.

During network startup or after a topology change, STP exchanges **Bridge Protocol Data Units (BPDUs)** between switches to determine the active topology. During this convergence period, the **first ICMP packet** generated by the ping command may be dropped. After convergence is complete, the forwarding ports begin transmitting traffic normally, and subsequent ping requests are successful.

Resolution

No corrective action is required because the network is operating correctly. The **Blocking (BLK)** state is automatically maintained by STP to prevent Layer 2 switching loops. If one of the active forwarding links fails, STP recalculates the topology and transitions the blocked port into the **Forwarding** state, ensuring continuous network connectivity without manual intervention.

Verification

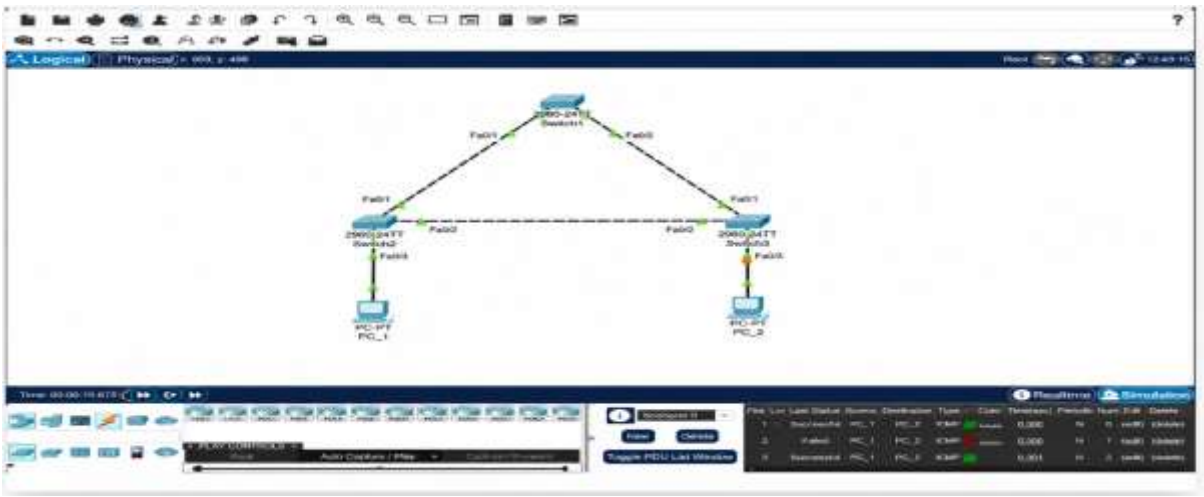
1. Open the switch CLI.
2. Execute:

Switch# show spanning-tree

3. Verify that:
    - One interface is in the **BLK (Blocking)** state.
    - Remaining interfaces are in the **FWD (Forwarding)** state.
    - The Root Bridge information is displayed.
    - The Root Port and Designated Ports are correctly identified.
4. Switch to **Simulation Mode** in Cisco Packet Tracer.
5. Filter **STP** and **ICMP** packets.
6. Observe the exchange of **BPDUs** between the switches.
7. Ping **PC_2** from **PC_1**. The first ping may time out during STP convergence, while the remaining ping packets are transmitted successfully after convergence.

Conclusion

The **amber port in the BLK (Blocking) state** represents the normal operation of **Spanning Tree Protocol** in a redundant switched network. The show spanning-tree command confirms that STP has selected a loop-free topology by assigning port roles and blocking one redundant path. This mechanism prevents **Layer 2 loops**, **broadcast storms**, **duplicate frame transmission**, and **MAC address table instability**, while keeping the blocked link available as a backup path in case the active forwarding link fails.



Part 2: Optimization Scenarios

6. Root Bridge Optimization

- **Scenario: A large campus network experiences suboptimal path choices. Layer 2 traffic traveling between two local departments is routed through a legacy, low-spec edge switch instead of the core distribution switch.**
- **Packet Tracer Check: Run show spanning-tree on all switches to find which device says This bridge is the root. You find that an old switch won the root election because it happens to have the lowest burned-in MAC address.**
- **Optimization Question: What structural problem does an unoptimized Root Bridge create for STP path calculations? Provide the Cisco IOS configuration commands to manually force your primary Core Switch to always be elected the Root Bridge for VLANs 10, 20, and 30.**

7. EtherChannel Link Aggregation

- **Scenario: The trunk link between two core switches is constantly saturated at 100% bandwidth capacity during peak operational hours, creating a network performance bottleneck.**
- **Packet Tracer Check: In Simulation Mode, packets queue up heavily on a single interface. You decide to bundle two physical FastEthernet ports (F0/23 and F0/24) into a single logical channel.**
- **Optimization Question: Configure an active LACP EtherChannel bundle between the two switches. Write out the interface configuration steps. How does this aggregation optimize layer 2 throughput and provide fault tolerance if one physical cable is disconnected?**

8. PortFast and BPDU Guard Implementation [1]

- **Scenario: When user workstations boot up or reboot, it takes up to 30–50 seconds before they can pull an IP address from the DHCP server or pass network traffic.**
- **Packet Tracer Check: Watch the link light turn from amber to green over nearly a minute during a device power cycle.**
- **Optimization Question: Which Data Link layer sub-protocol causes this transition delay? Write down the optimization command applied to edge access ports to bypass this delay safely. What security command should be paired with it to prevent unauthorized switches from being plugged into those access ports?**

6.Root Bridge Optimization

Structural Problem

In this campus network, the **legacy edge switch** has been elected as the **Root Bridge** because it has the lowest burned-in MAC address. Although STP automatically selects the switch with the lowest Bridge ID, this may not always be the most suitable device for the Root Bridge role. Since the Root Bridge acts as the central reference point for all STP path calculations, every switch determines its forwarding path based on the shortest path to this device.

When an edge switch becomes the Root Bridge, traffic between local departments is forced to pass through that switch instead of the high-performance core distribution switch. This results in inefficient traffic forwarding, longer communication paths, increased latency, and unnecessary utilization of lower-capacity links. The core switch, which is designed to handle large volumes of traffic, remains underutilized while the edge switch experiences additional processing and forwarding load. As the network grows, this poor Root Bridge placement can reduce overall network performance and increase congestion.

Command Used to Identify the Root Bridge

Run the following command on every switch:

Switch# show spanning-tree

The switch displaying the following message is currently acting as the Root Bridge:

This bridge is the root

The command also displays the Root Bridge ID, bridge priority, MAC address, interface roles, and STP port states.

Cisco IOS Configuration

Configure the Core Switch as the primary Root Bridge for VLANs 10, 20, and 30.

CoreSwitch(config)# spanning-tree vlan 10 root primary

CoreSwitch(config)# spanning-tree vlan 20 root primary

CoreSwitch(config)# spanning-tree vlan 30 root primary

Alternatively, assign a lower bridge priority manually.

CoreSwitch(config)# spanning-tree vlan 10 priority 4096

CoreSwitch(config)# spanning-tree vlan 20 priority 4096

CoreSwitch(config)# spanning-tree vlan 30 priority 4096

Verification

Run:

CoreSwitch# show spanning-tree

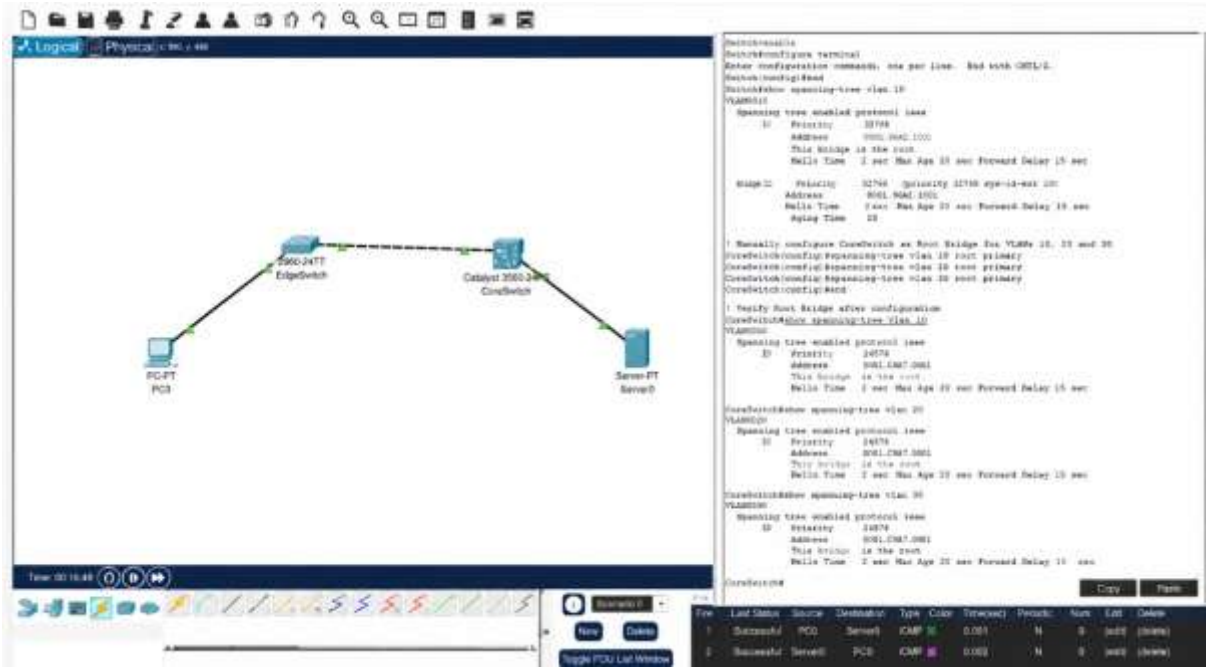
Verify that the output displays:

This bridge is the root

for VLANs **10, 20, and 30**.

Conclusion

After configuring the Core Switch as the Root Bridge, STP recalculates the network topology and selects forwarding paths through the central switch. This provides shorter communication paths, reduces unnecessary traffic through edge switches, improves bandwidth utilization, minimizes forwarding delays, and creates a more efficient Layer 2 topology. The optimized Root Bridge placement ensures better scalability, higher performance, and more stable network operation.



7. EtherChannel Link Aggregation

Network Issue

The trunk link connecting the two core switches is operating at **100% bandwidth utilization** during peak traffic periods. All data traffic is transmitted through a single FastEthernet interface, creating a network bottleneck. As the number of users and applications increases, packets begin to queue on the interface, increasing transmission delay and reducing network performance. This single-link dependency also creates a risk of network interruption if the physical cable fails.

To improve bandwidth utilization and provide redundancy, the two FastEthernet interfaces (**F0/23** and **F0/24**) can be combined into a single logical interface using **EtherChannel** with the **Link Aggregation Control Protocol (LACP)**.

Configuration Steps

Switch 1

Switch1(config)# interface range fastEthernet0/23 - 24

Switch1(config-if-range)# switchport mode trunk

Switch1(config-if-range)# channel-group 1 mode active

Switch1(config-if-range)# exit

Switch1(config)# interface port-channel 1

Switch1(config-if)# switchport mode trunk

Switch1(config-if)# no shutdown

Switch 2

Switch2(config)# interface range fastEthernet0/23 - 24

Switch2(config-if-range)# switchport mode trunk

Switch2(config-if-range)# channel-group 1 mode active

Switch2(config-if-range)# exit

Switch2(config)# interface port-channel 1

Switch2(config-if)# switchport mode trunk

Switch2(config-if)# no shutdown

Verification

Execute:

Switch# show etherchannel summary

The output should indicate that **Port-Channel 1** is operational and that interfaces **F0/23** and **F0/24** are successfully grouped into the EtherChannel.

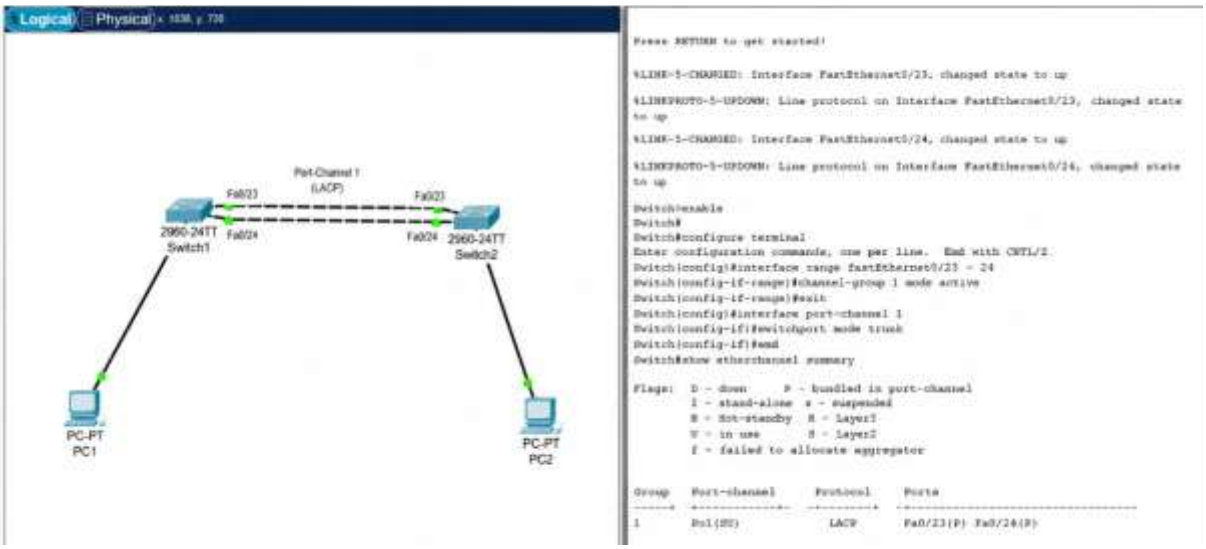
Optimization

EtherChannel combines multiple physical interfaces into one logical interface, allowing traffic to be distributed across all available links. Instead of transmitting all frames through a single interface, the switch balances traffic across both links using a load-balancing algorithm. This effectively increases the available bandwidth and reduces interface congestion.

EtherChannel also provides fault tolerance. If one physical cable or interface fails, the remaining active interface continues forwarding traffic through the Port-Channel without requiring STP to recalculate the topology. This minimizes downtime and ensures uninterrupted communication between the switches.

Conclusion

The implementation of LACP EtherChannel increases the available bandwidth between the core switches, distributes traffic efficiently, reduces network congestion, and provides automatic redundancy. The logical Port-Channel simplifies network management while improving overall Layer 2 performance and reliability.



8. PortFast and BPDU Guard Implementation

Network Issue

User workstations connected to the access switches require approximately **30 to 50 seconds** before they receive an IP address from the DHCP server or begin transmitting network traffic. During device startup, the switch port initially displays an amber light before turning green. This delay affects users by increasing workstation startup time and delaying access to network resources.

The delay occurs because **Spanning Tree Protocol (STP)** places every newly connected switch port through several initialization states before allowing normal data forwarding.

Data Link Layer Sub-Protocol

The transition delay is caused by the **Spanning Tree Protocol (STP)**.

During initialization, the port progresses through the following states:

- Blocking
- Listening
- Learning
- Forwarding

Only after reaching the **Forwarding** state can the workstation communicate with the DHCP server and other network devices.

PortFast Configuration

Enable PortFast on access ports connected only to end devices.

Switch(config)# interface fastEthernet0/1

Switch(config-if)# switchport mode access

Switch(config-if)# spanning-tree portfast

To enable PortFast on all access interfaces:

Switch(config)# spanning-tree portfast default

BPDU Guard Configuration

Enable BPDU Guard to secure PortFast interfaces.

Switch(config)# interface fastEthernet0/1

Switch(config-if)# spanning-tree bpduguard enable

Or enable it globally:

Switch(config)# spanning-tree portfast bpduguard default

Verification

Run the following command:

Switch# show spanning-tree interface fastEthernet0/1 detail

Verify that the interface has **PortFast enabled** and is operating normally.

Conclusion

PortFast allows access ports connected to user devices to transition immediately to the **Forwarding** state, eliminating the normal STP startup delay. As a result, workstations obtain IP addresses more quickly and network communication begins immediately after booting. BPDU Guard enhances network security by automatically disabling a PortFast-enabled port if BPDU packets are received, preventing unauthorized switches from being connected and protecting the Layer 2 topology from accidental or malicious changes. Together, PortFast and BPDU Guard improve network performance, reduce connection delays, and maintain a secure and stable switching environment.

